10TH CLASS COMPUTER SCIENCE — NEW SYLLABUS (PECTAA)

Chapter 7 — Applications of AI

Solved Exercise | New Book (PECTAA)

Exercise (Solved)

A. Multiple Choice Questions

1. What does AI stand for?

(a) Automatic Intelligence (b) Artificial Intelligence
(c) Active Interface (d) Applied Information

Answer: (b) — AI stands for Artificial Intelligence. It is the field of computer science in which machines are made to “think” and “act” like humans, that is, to solve problems, learn from the data and take smart decisions.

2. Which of the following is an example of Natural Language Processing (NLP)?

(a) Online calculator (b) Voice typing
(c) File copying (d) Image editing

Answer: (b) — NLP is the part of AI which helps the computer to understand the human language. Voice typing converts the spoken words into written text, so it uses NLP. An online calculator, copying a file and editing an image do not deal with the human language.

3. What type of data helps AI make better decisions?

(a) Short and weak data (b) Only images
(c) Large and good quality data (d) Data without labels

Answer: (c) — AI learns from the data in the same way as a student learns from books. When the data is large in quantity and correct in quality, the AI system becomes more familiar with the different situations and gives better and more reliable results.

4. Which tool uses AI to understand human voice?

(a) Video player (b) Translator
(c) Speech recognition (d) Mouse

Answer: (c) — Speech recognition is the AI technology which listens to the spoken words and turns them into text, so that the machine can understand and answer. Voice assistants in phones and smart devices work with this technology.

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5. What is the role of robots in factories?

(a) Watching videos (b) Making websites

(c) Doing repeated or dangerous tasks (d) Writing stories

Answer: (c) — In factories the robots do the repeated and the dangerous tasks, such as lifting heavy items, welding and painting. They do not get tired and they work with great accuracy, so the work becomes faster and safer.

6. What is a challenge of using AI?

(a) It works slowly (b) It cannot be turned off

(c) It may replace human jobs (d) It cannot store data

Answer: (c) — One big challenge of AI is that it can replace the human jobs in some fields, such as customer service, data entry and clerical work, which may lead to job loss. Other challenges are the wrong answers due to the poor or biased data.

7. If data represents only a group of people it is called:

(a) Unbiased data (b) Biased data

(c) Good data (d) Reliable data

Answer: (b) — When the data represents only one group of people, it is called biased data. If the AI is trained only on such data, it may not understand the other groups fairly; for example, a face recognition tool trained mostly on one type of images may not work well for other people.

8. Why is fairness important in AI?

(a) So the system looks good (b) To avoid slow speed

(c) To give equal treatment to all users (d) To use more memory

Answer: (c) — Fairness in AI means that the system treats all the users equally, no matter what their race, gender or background is. This is very important in the systems which take the decisions about the loans, the school admissions, the jobs and the legal cases.

9. Which of these is NOT an AI tool?

(a) Chatbot (b) Face detection

(c) Spell checker (d) Paper notebook

Answer: (d) — A paper notebook is an ordinary stationery item; it cannot learn, understand or take any decision. A chatbot, face detection and a spell checker all use AI, because they work with the human language or with the images.

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B. Short Questions

1. What is Artificial Intelligence (AI)?

Artificial Intelligence is the field of computer science in which machines are taught to “think” and “act” like humans. It enables the computers to solve the problems, to learn from the data and to take smart decisions without being given step-by-step instructions. In this way the machines perform the tasks which normally need human intelligence, such as understanding the language, recognising the faces, driving a car and answering the questions of the customers.

2. How is AI used in daily life?

AI is used in our daily life in many ways, often without our notice. It suggests the posts and the videos we may like on Facebook, YouTube and Instagram, and the products we may buy on the online stores. It works in the voice assistants of our phones, in the face unlock of the smartphones and in the route suggestions of Google Maps. In the hospitals it helps in the diagnosis, in the banks and the telecom companies it answers the customers through the chatbots, and in the schools it guides the students with the smart learning apps.

3. Why does AI need large and good-quality data?

AI learns from the data, just as a student learns by reading the books, so the quantity and the quality of the data both matter. When the AI receives a large number of examples, it becomes familiar with the different situations and can handle a new case correctly. If the data is of good quality, that is, correct and free from mistakes, the AI learns the right things. But if the data is small, wrong or biased, the AI gives the wrong answers and takes the unfair decisions.

4. What is the role of Natural Language Processing (NLP)?

Natural Language Processing is the part of AI which helps the computers to understand the human language. It lets a machine read the text, listen to the speech and give the answer in a way which the people can understand. Because of NLP, a chatbot can reply to the questions of the customers 24 hours a day, a search engine can understand what you are looking for, and a voice assistant can do the work which is asked by speaking. It is also used in the automatic translation apps and in the text-analysis tools.

5. Give one example of robotics in everyday life.

A good example is the **robot vacuum cleaner** used in the homes. It moves around the room by itself, detects the walls and the furniture with its sensors and cleans the floor intelligently, without any person to guide it. Robotics is also used in the hospitals for surgeries, in the factories for lifting the heavy items and in the space missions.

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6. What is the purpose of speech recognition technology?

Speech recognition is an AI technology which allows the computer to listen and understand the spoken words and to turn them into text. Its purpose is to make it easy and fast for the people to use the machines with their voice only, instead of typing or pressing the buttons. For example, the voice assistants in the phones and the smart devices follow the spoken commands, the smart homes turn the lights on or off by voice, and the doctors record their notes by speaking without typing.

7. How do recommendation systems work?

Recommendation systems are AI programs which suggest the videos, the audios or the products according to the likes of the user. They study the data of the user, such as what he has watched, searched or bought before, and compare it with the data of the other users who have similar habits. In this way they find the items which the user will probably like and show them on the screen; for example, YouTube suggests the videos similar to the ones you have already watched, and an online store suggests the products related to your previous purchases.

8. What is meant by bias in AI systems?

Bias in AI means that the system gives unfair results because of the data on which it was trained. If the training data represents only one group of people, the AI cannot understand the other groups fairly and its decisions go in favour of one side. For example, a face recognition tool trained mostly on the images of one type of people may not work well for the people of the other types. That is why the training data must include different people, languages and situations.

9. Why should AI systems be transparent and fair?

Fairness means that the system treats all the users equally, no matter what their race, gender or background is, and **transparency** means that the people can understand how the system works and why it took a certain decision. Both are necessary because the AI takes the decisions about the loans, the school admissions, the jobs and the legal cases, and if such decisions are unfair or hidden, they can harm the people. When an AI rejects a loan application, the user has the right to know which rules were followed, so that he can be treated justly.

C. Long Questions

1. What is the importance of data in AI? Explain why good and large data is needed.

Data is the most important part of AI, because an AI system learns from the data in the same way as a student learns by reading books. The model is trained on the old examples, and from them it learns the patterns which it uses later to make predictions and decisions. Without data there is nothing from which the machine can learn, so the whole working of AI depends on data.

(i) Quantity of data: when the AI receives a large number of examples, it becomes familiar with the many different situations. For instance, if a machine sees a very large number of images or texts, it becomes better at understanding a new image or a new text which it has never seen.

(ii) Quality of data: the quality means how correct and useful the data is. If the data has mistakes, missing values or confusing labels, the AI learns the wrong things and gives the wrong answers. A model trained on poor-quality data may even make unfair decisions, for example a face recognition system trained only on one type of images.

(iii) Accuracy of the decisions: the quantity and the quality together decide how reliable the AI is. Good and large data gives the correct patterns, so the predictions of the model become accurate; poor data makes the model useless, however good the algorithm may be.

That is why the data is collected automatically in a large quantity from the phones, the websites and the apps, and it is cleaned and checked before the training. In short, an AI learns better only with the big (quantity) and the accurate (quality) data.

2. Write in detail about any three real-world applications of AI (such as NLP, robotics, or speech technology).

(a) Natural Language Processing (NLP): NLP is the part of AI which helps the computers to understand the human language. It allows a machine to read the text, to listen to the speech and to answer in a way which the people can understand. When someone types a question in the search bar or talks to a voice assistant, NLP helps the system to understand the question and to reply correctly. In the customer service, the chatbots on the websites answer the common questions 24 hours a day with the help of NLP, and the automatic translation apps change one language into another. In this way NLP is used in the search engines, in the chatbots and in the text-analysis tools.

(b) Robotics: robotics is the sub-field of AI which involves the machines designed to do tasks like humans; these machines are called robots. A robot can lift heavy items, help in the surgeries, clean the floors intelligently and help the old or the disabled people in their daily tasks. Because the robots do not get tired and they work with great accuracy, they are used in the factories for welding, painting and assembling the parts, in the hospitals, in the agriculture and even in the space missions. In everyday life a robot vacuum cleaner is a common example.

(c) Speech recognition and speech technology: speech technology is the type of AI which allows the computers to listen and understand the spoken words. It turns the voice into text and responds with an answer. The voice assistants in the phones and the smart speakers work with it; the smart homes turn the lights on or off by a voice command; in the customer service it answers the calls and guides the users without any human operator; and in the healthcare it helps the doctors to record their notes without typing. Some cars also understand the voice commands to make calls or to turn on the music.

All these applications save the time and the effort of the people, reduce the human errors and increase the efficiency of the work, but they also need the good-quality data and the careful design, so that they remain fair and safe.

3. Explain the ethical issues in AI. Why are fairness, privacy, and transparency important?

Ethics in AI means using AI in a way which is fair, honest and safe for everyone. Because AI takes decisions which affect the people, it is necessary that these decisions do not hurt anyone or treat any group unfairly. The main ethical issues are the following.

(i) Bias: bias in AI happens when the system gives unfair results because of the data on which it was trained. If the data represents only one group of people, the AI may not understand the other groups fairly; for example, a face recognition tool trained mostly on the images of one type of people may not work well for the others.

(ii) Privacy and data security: AI systems collect a large amount of personal data such as the names, addresses, locations and search history. If this data is not protected, it can be stolen or misused by the hackers, so the personal information must be kept safe and must not be shared without permission.

(iii) Surveillance: the monitoring systems and the cameras watch the actions of the people. This can be useful for the security, but watching the people all the time without any reason harms their freedom and privacy.

(iv) Social equity: sometimes the new AI tools are available only to the rich or the educated groups, while the poor, the disabled and the people of the remote areas are left behind, and this gap between the people increases.

(v) Responsibility of the designers: the AI systems do not make the rules by themselves; they follow the instructions of the human designers. Therefore the people who build the AI have a big responsibility to make it fair, safe and helpful.

Importance of fairness: fairness means that the system treats all the users equally, no matter what their race, gender or background is. It is important because the AI takes the decisions about the loans, the school admissions, the job interviews and the legal cases, and an unfair system can destroy the life of a person. To ensure fairness, the AI must be trained on the data which includes all types of people and situations.

Importance of privacy: privacy means keeping the personal information safe and not sharing it without permission. It is important because the users give their personal data to the AI systems, and if it is not protected, it can be stolen or misused. The protection of the data keeps the trust of the people in these systems.

Importance of transparency: transparency means that the people should understand how an AI system works and why it took a certain decision. It is important because if an AI rejects a loan application or does not select a student, the affected person has the right to know which rules and steps the system followed. Transparency makes the system accountable and removes the doubt of unfairness.

In short, the AI must be developed with care, keeping in mind fairness, privacy, transparency and the respect for human rights, so that it benefits all the people equally and no one is harmed.